Title: Comprehensive Network Reconnaissance & Attack Surface Mapping

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Course: BCA (2nd Year)

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Domain: Cyber Security (Minor Project)

1. Objective:

The objective of this project is to perform network reconnaissance in a controlled environment to identify active hosts, open ports, running services, and potential vulnerabilities. The project also aims to analyse the network traffic and understand the attack surface of a target system.

2. Introduction:

In Cybersecurity, reconnaissance is the first and the most important phase of ethical hacking. It involves gathering information about a target system or network before attempting any attack. Attackers use reconnaissance to identify weak points such as open ports, running services, and exposed systems.

This project demonstrates how ethical hackers use tools to scan networks, analyse traffic, and identify potential security risks in a safe and controlled environment.

3. Tools Used:

- **Kali Linux**
- **Nmap**
- **Netdiscover**
- **Wireshark**

4. Methodology:

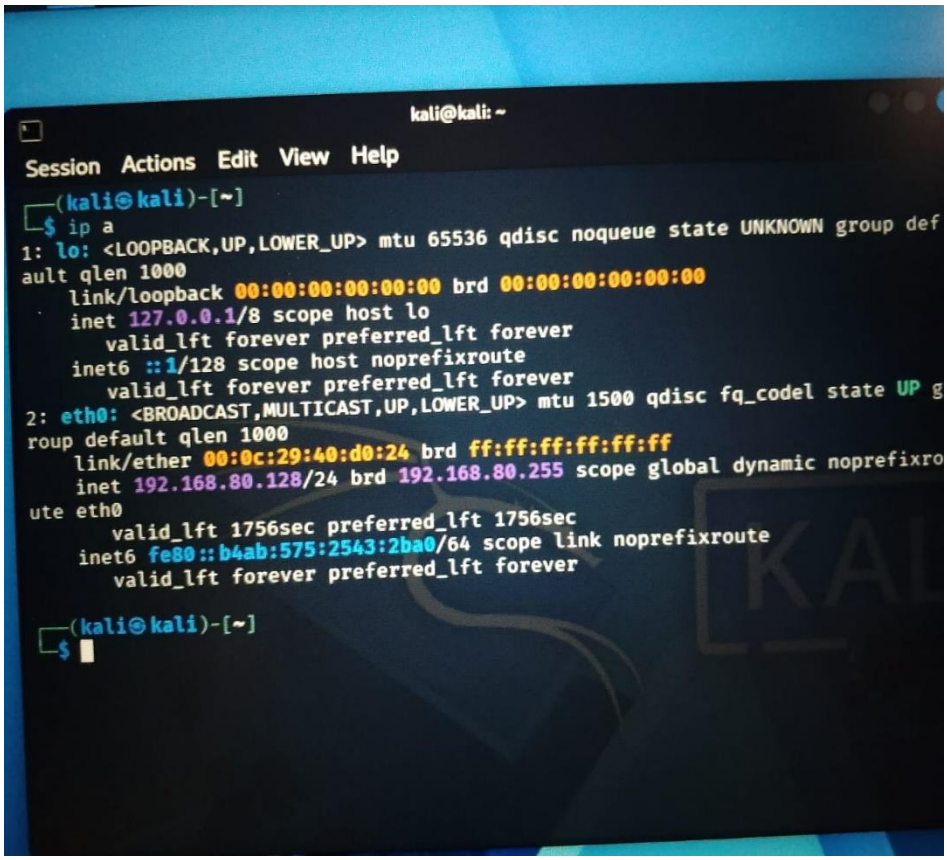
- **Design a reconnaissance strategy based on ethical hacking methodology**
- **Perform host discovery and Service enumeration**
- **Identify OS Fingerprints and Running services**
- **Capture and Analyse Network packets for unusual Behaviour**
- **Create a Detailed Attack Surface**

5. Reconnaissance Strategy:

The Reconnaissance Strategy is followed by a structural ethical hacking methodology. Initially, the attacker system was identified, by the network host discovery using automated tools. A targeted system was then selected for detailed scanning. Progressive enumeration techniques were applied to identify open ports, services, and operating system details. Finally, traffic analysis was performed to observe the network behaviour and validate findings.

6. Practical Steps:

Step 1: IP Address Identification



Command Used: <u>ip a</u>

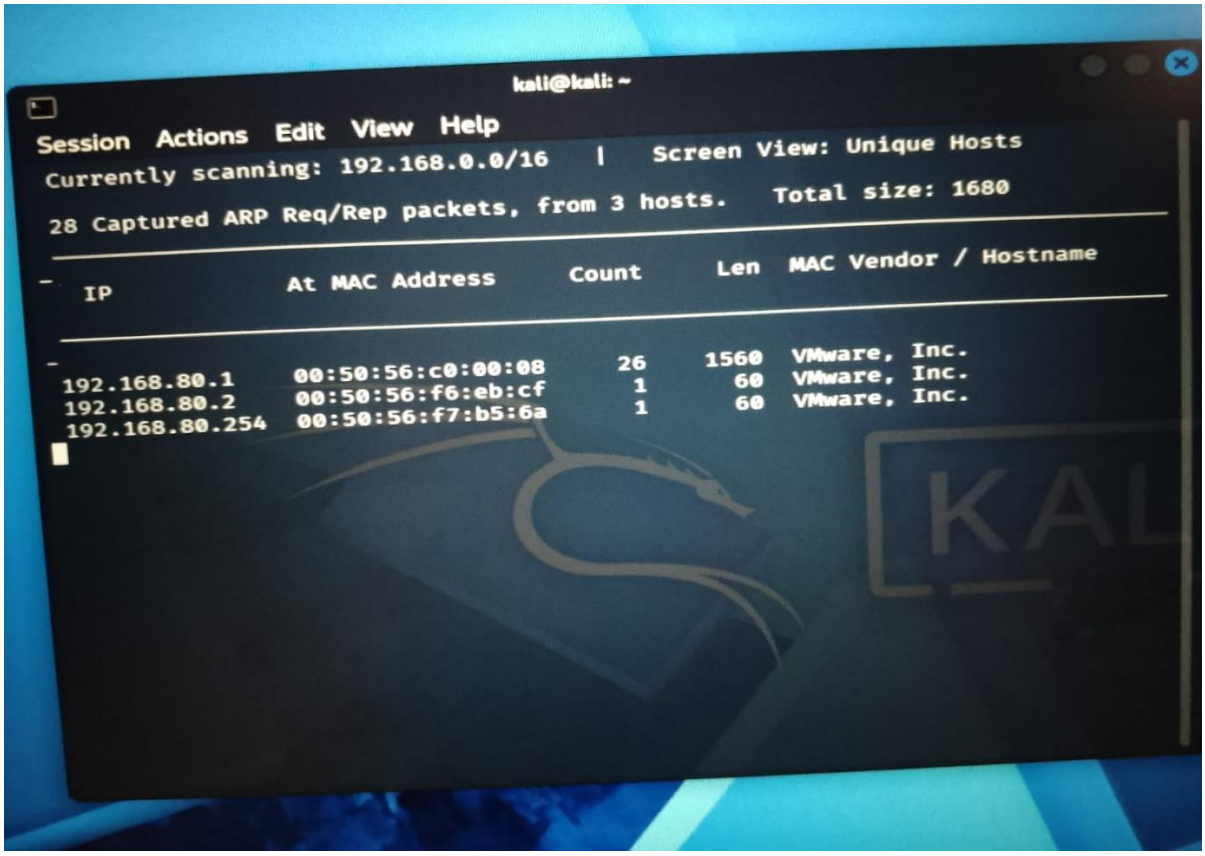
The <u>ip a</u> command was used to identify the IP address of the attacker machine. This step is important to understand the system's position in the network before performing reconnaissance.

Result

The IP address identified is:

- **192.168.80.128**

Step 2: Host Discovery



Command Used: sudo netdiscover

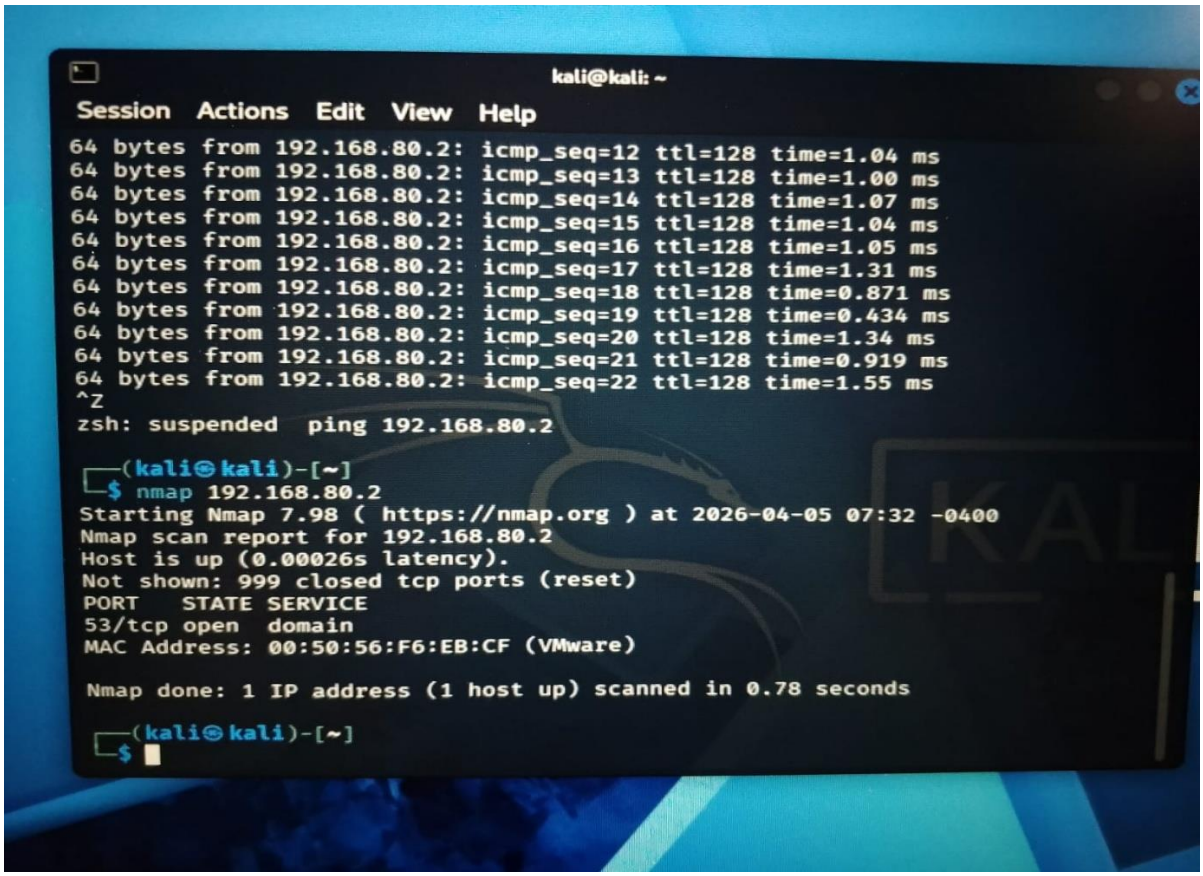
Netdiscover was used to scan the local network and identify active hosts. It provides information such as IP addresses, MAC addresses, and vendor of connected devices.

Result:

Following are the active hosts were discovered:

- **192.168.80.1**
- **192.168.80.2**
- **192.168.80.254**

Step 3: Basic Scan using Nmap



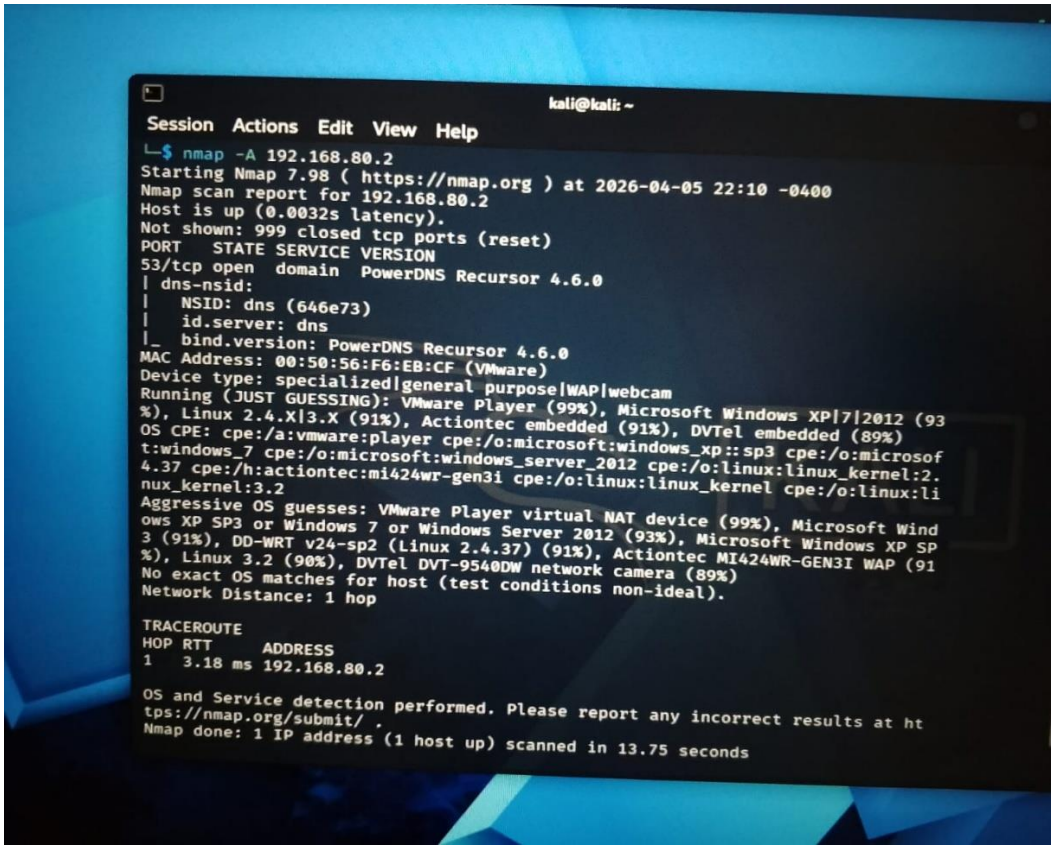
Command Used: nmap 192.168.80.2

A basic Nmap scan was performed on the selected target IP to identify open ports and services running on the system.

Result:

- **Host is active.**
- **Open Port: 53.**
- **Service: DNS (Domain Name System).**
- **All other ports are closed.**

Step 4: Advanced Scan using Nmap



```
kali@kali: ~
Session  Actions  Edit  View  Help
└─$ nmap -A 192.168.80.2
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-05 22:10 -0400
Nmap scan report for 192.168.80.2
Host is up (0.0032s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
53/tcp open  domain  PowerDNS Recursor 4.6.0
| dns-nsid:
|   NSID: dns (646e73)
|   id.server: dns
|_  bind.version: PowerDNS Recursor 4.6.0
MAC Address: 00:50:56:F6:EB:CF (VMware)
Device type: specialized|general purpose|WAP|webcam
Running (JUST GUESSING): VMware Player (99%), Microsoft Windows XP|7|2012 (93
%), Linux 2.4.X|3.X (91%), Actiontec embedded (91%), DVTel embedded (89%)
OS CPE: cpe:/a:vmware:player cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsof
t:windows_7 cpe:/o:microsoft:windows_server_2012 cpe:/o:linux:linux_kernel:2.
4.37 cpe:/h:actiontec:mi424wr-gen3i cpe:/o:linux:linux_kernel cpe:/o:linux:li
nux_kernel:3.2
Aggressive OS guesses: VMware Player virtual NAT device (99%), Microsoft Wind
ows XP SP3 or Windows 7 or Windows Server 2012 (93%), Microsoft Windows XP SP
3 (91%), DD-WRT v24-sp2 (Linux 2.4.37) (91%), Actiontec MI424WR-GEN3I WAP (91
%), Linux 3.2 (90%), DVTel DVT-9540DW network camera (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   3.18 ms 192.168.80.2

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.75 seconds
```

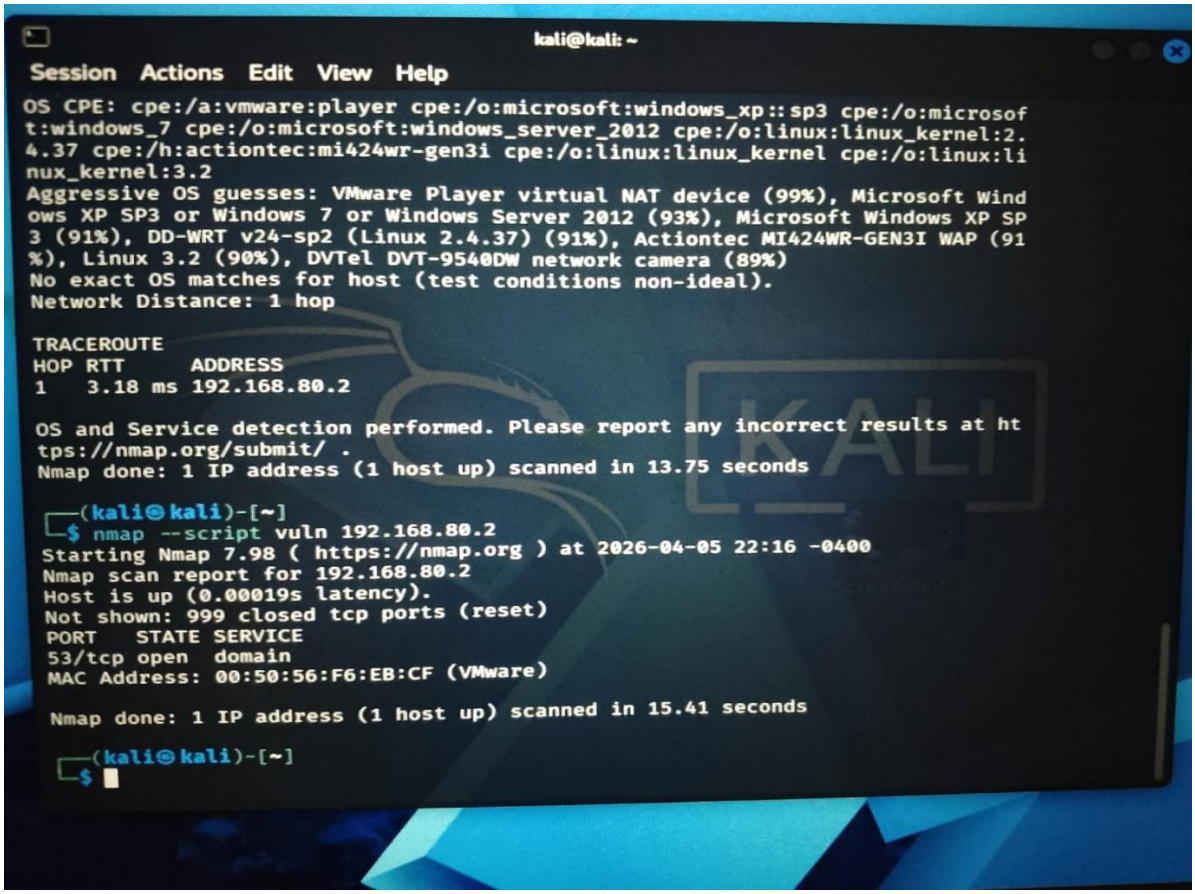
Command Used: nmap -A 192.168.80.2

An advanced scan was performed to gather detailed information such as service versions, operating system, and network details.

Result:

- **Port 53 is open.**
- **Service running: PowerDNS Recursor 4.6.0.**
- **OS detection: VMware virtual NAT device.**
- **Network distance: 1 hop.**

Step 5: Vulnerability Scan using Nmap



Command Used: nmap –script vuln 192.168.80.2

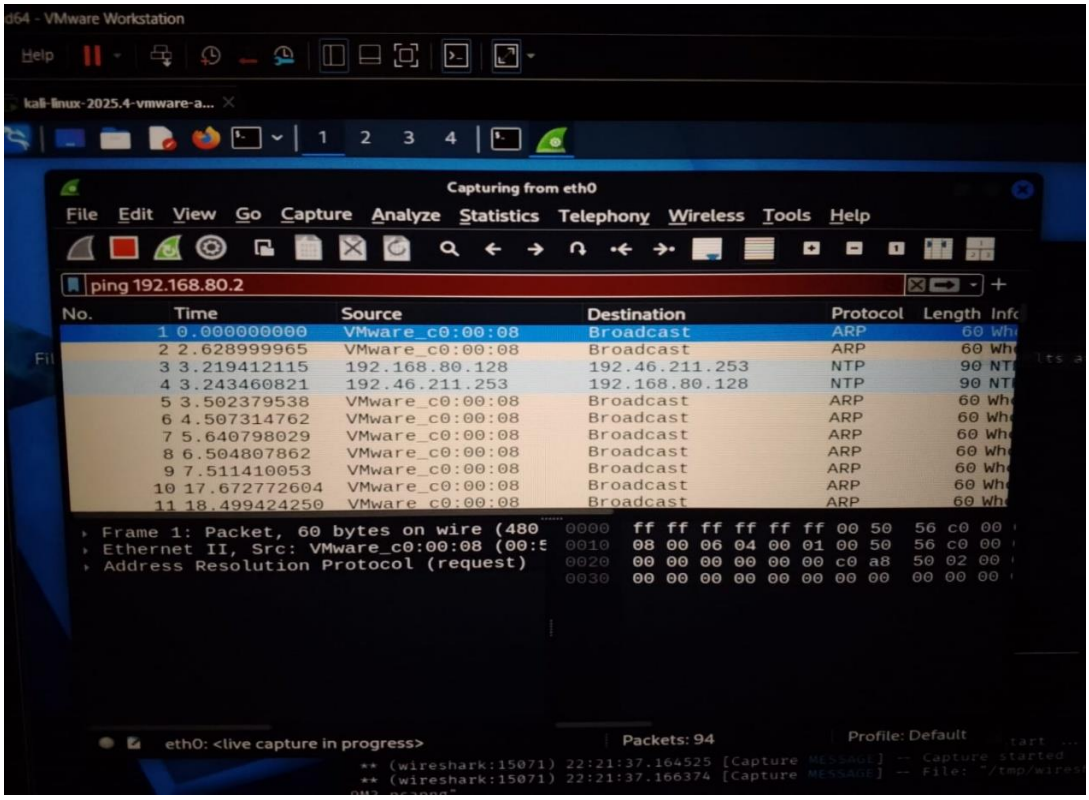
Nmap vulnerability scripts were used to check for known security vulnerabilities in the target system.

Results:

- **No vulnerabilities detected.**
- **Only port 53 (DNS) is open.**

This indicates the system is relatively secure against common vulnerabilities.

Step 6: Traffic Analysis using Wireshark



Tool Used: Wireshark

Wireshark was used to capture and analyse network traffic. It helps in identifying communication between devices and detecting unusual behaviour.

Results:

- **Protocols observed: ARP and NTP.**
- **ARP shows communication within local network.**
- **NTP is used for time synchronization.**
- **No suspicious or malicious activity detected.**

External Link:

[AaryanYT/Cybersecurity-Project]